

Problem Statement :

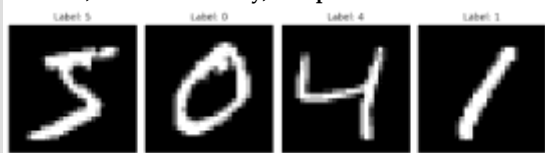
Multiclass classification in CNN commonly uses One vs Rest (OvR) and One vs One (OvO) strategies, yet their trade offs in accuracy, computational cost, and prediction stability across modern architectures are not well quantified. This work empirically compares OvR and OvO on LeNet, MobileNetV2, and ResNet18 using the MNIST dataset

Objectives :

- To compare OvR and OvO multiclass strategies across CNN backbones of varying depth and efficiency.
- To evaluate their performance in terms of accuracy, macro F1-score, and inference latency on the MNIST dataset.
- To analyze per class prediction stability and quantify the computational overhead of each strategy.
- To identify the most efficient multiclass approach for modern CNN based classification.

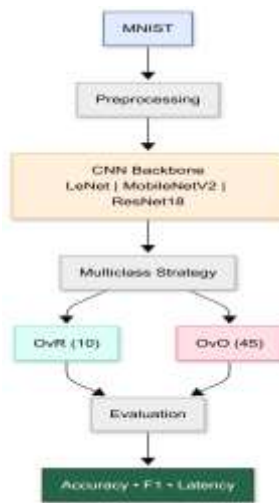
Methodology :

The MNIST dataset was used with standard train-test splits. Three CNN backbones like LeNet, MobileNetV2, and ResNet18 were evaluated to represent. Multiclass classification was implemented using OvR and OvO strategies, requiring 10 and 45 binary classifiers respectively. Each binary classifier was trained using binary cross entropy loss. Performance was evaluated using accuracy, macro F1-score, inference latency, and per-class behavior.



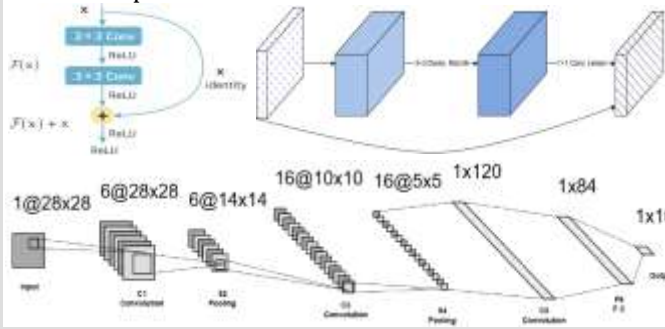
Proposed work :

Implement OvR and OvO multiclass strategies on LeNet, MobileNetV2, and ResNet18. Train the complete set of 165 binary classifiers under controlled and reproducible experimental settings. LeNet is trained from scratch, MobileNetV2 uses a frozen pretrained backbone, and ResNet18 is fine tuned. Perform inference using maximum score selection for OvR and majority voting for OvO. Evaluate models using accuracy, macro F1-score, and inference latency, along with per-class performance analysis. Compare computational overhead and prediction stability between OvR and OvO across architectures.



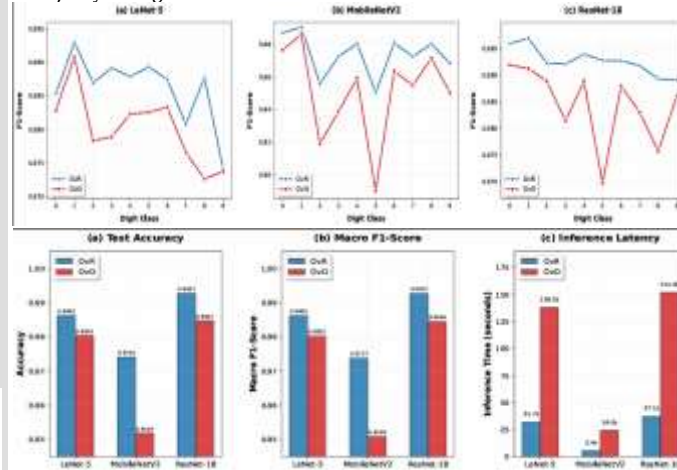
Architecture :

LeNet: used as a classical baseline with limited feature capacity.
MobileNetV2: Lightweight pretrained CNN with a frozen backbone.
ResNet18: Deep residual CNN fine-tuned end to end.



Results & Discussions :

OvO got 4 to 5 times higher inference latency due to pairwise classifiers, particularly affecting lightweight models. ResNet18 with OvR achieved the highest accuracy (99.27%). Per-class analysis showed more stable predictions with OvR, while OvO exhibited higher variability from majority voting.



Conclusion : OvR consistently matches or outperforms OvO across all evaluated CNN backbones while requiring significantly lower computational cost. OvO introduces higher inference latency and prediction instability due to pairwise voting, particularly in lightweight architectures. Strong feature extractors such as ResNet18 enable OvR to achieve the best performance, making complex multiclass decomposition unnecessary.

Reference :

- R. Rifkin and A. Klautau, "In Defense of One-vs-All Classification," Journal of Machine Learning Research, vol. 5, pp. 101-141, 2004.
K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778.